EEEM068: Applied Machine Learning

Diabetic Retinopathy Grading Using Vision Transformers and Deep Learning

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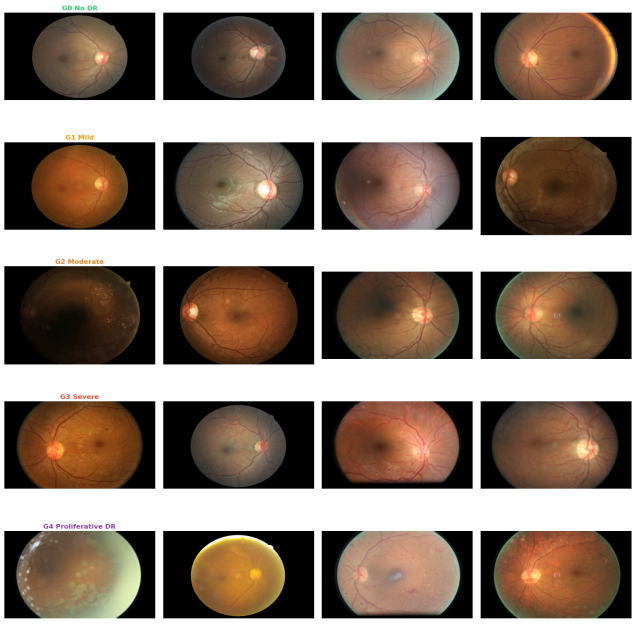
Abstract

Diabetic retinopathy, a retinal disease linked to diabetes, requires early detection and severity grading to prevent vision loss. This project applies machine learning to a real medical image classification problem using digital colour fundus photography from the Kaggle Diabetic Retinopathy Detection dataset. The task is challenging due to varying image quality and dataset imbalance, with most images belonging to the No DR class. To investigate this, we developed a complete PyTorch based pipeline and compared an EfficientNet baseline with advanced deep learning architectures, including transformer-based and self-supervised vision models. The pipeline included image preprocessing, data augmentation, class imbalance handling, model fine-tuning, validation, and testing. Models were evaluated using accuracy, precision, recall, F1-score, balanced accuracy, confusion matrix analysis, and quadratic weighted kappa. The best performing model was achieved. However, minority class recognition, especially Mild DR, remained difficult due to subtle visual differences and limited class representation. Overall, the project demonstrates how machine learning can support medical image analysis, emphasising the importance of robust evaluation, class imbalance handling, and clinically meaningful error analysis.

I. INTRODUCTION

Diabetic retinopathy (DR) is the leading cause of blindness among working-age people around the world. Preventable since it is a degenerative disease caused by prolonged high blood glucose levels that damage the eye's microvascular structures. The extent of this disease ranges from having no signs of retinopathy to proliferative DR requiring urgent intervention. The existing technique for screening uses expert ophthalmologists who manually examine color images of the eye's fundus.

Convolutional neural networks and vision transformers, two recent developments in deep learning, have demonstrated exceptional promise in automating DR identification and grading. The Kaggle Diabetic Retinopathy Detection competition catalysed much of this research, providing a large-scale labelled dataset of retinal images graded on a five-point severity scale (0-4). However, the dataset exhibits extreme class imbalance, with approximately 73% of images belonging to the healthy class (Grade 0), posing a significant challenge for model training and evaluation. While early approaches relied heavily on CNN architectures such as ResNet and EfficientNet, the emergence of new paradigms for capturing long-range spatial dependencies in medical images. This is particularly relevant to DR grading, where clinically significant features range from tiny microaneurysms and dot haemorrhages to large scale exudates and neovascularisation, suggesting that the global attention mechanism of transformers may capture these multi-scale patterns more effectively than the local



receptive fields of CNNs.

In this project, we conduct a systematic comparative study of five deep learning architectures for DR grading. Each group member developed and optimised a distinct model ViT-B/16, Swin Transformer, DeiT III, DINOv2 (ViT-B/14), and MobileNetV3-Large, with EfficientNet serving as the shared baseline. These models were selected to represent different architectural paradigms: standard and hierarchical supervised transformers, self-supervised transformers with strong pre-trained representations, and

lightweight CNNs suited for resource-constrained deployment. Our objective is to identify which paradigm best addresses the challenges of class imbalance, subtle inter-class differences, and imaging variability inherent to DR grading.

II. LITERATURE REVIEW

A. Deep Learning for Diabetic Retinopathy

Gulshan et al. [1] demonstrated that a deep CNN trained on a large dataset of retinal images to achieve ophthalmologist level sensitivity and specificity for detecting referable DR, marking a pivotal moment for AI-assisted screening. Their work validated the use of transfer learning from ImageNet pretrained models for medical image classification, a strategy widely adopted in subsequent studies.

Gargeya and Leng [2] improved automated DR detection by fusing data driven feature learning with deep residual networks. Their method demonstrated the importance of data augmentation and cautious class imbalance management via weighted sampling. Early CNN based techniques performed well when capturing local textural features, but it was challenging to model global spatial relationships due to their small receptive fields.

B. Vision Transformers in Medical Imaging

Dosovitskiy developed the Vision Transformer (ViT), which uses a typical transformer encoder to process images divided into fixed-size patches. ViT enables models to represent long-range dependencies between distant retinal regions by replacing global self-attention with local convolutions. This is crucial for DR, as diseased traits vary in scale and spatial distribution. However, pre-training with large datasets like JFT-300M limited ViT's availability for medical imaging researchers.

The Swin Transformer, developed by Liu et al. [4], uses shifted window attention and hierarchical feature maps to address the quadratic complexity of ViT. Its multi-scale feature representation is ideal for DR grading, where clinically important results range from tiny hemorrhages to extensive exudation or neovascularization. Touvron presented DeiT III, a data-efficient training recipe for ViTs that achieves competitive performance through sophisticated augmentation and optimization techniques, reducing reliance on extensive pre-training datasets [5].

Oquab introduced DINOv2, a series of self-supervised ViT models pre-trained on a carefully selected dataset of 142 million photos using self-distillation and masked image modeling [6]. These models successfully transfer highly generic visual cues to subsequent tasks., including medical image classification, often surpassing supervised pre-training approaches. Its self-supervised representations have shown particular resilience to the noise, illumination variation, and imaging artefacts common in retinal photography.

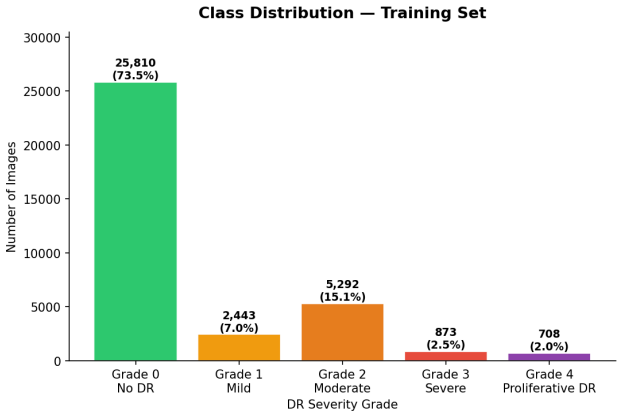
C. CNN Baselines and Efficiency

Tan and Le [7] introduced EfficientNet, which achieves better accuracy efficiency trade-offs by balancing network depth, width, and resolution through compound scaling. MobileNetV3, developed by Howard et al. [8], was optimized for mobile deployment using new activation functions and neural architecture search. These lightweight architectures continue to be significant as both rigorous baselines and viable options for resource constrained clinical application, where inference speed and model size are crucial factors, even as transformer based models push classification performance.

III. METHODOLOGY

A. Dataset Description

We utilise the Kaggle Diabetic Retinopathy Detection dataset, which contains high-resolution colour fundus



photographs captured under diverse imaging conditions. Each image is labelled with a DR severity grade from 0 (no DR) to 4 (proliferative DR). The dataset exhibits significant class imbalance, with the majority of images belonging to class 0.

B. Data Preprocessing and Augmentation

To match the distributions of the pre-trained models used for transfer learning, all retinal fundus images were scaled to 224 by 224 pixels and normalized using ImageNet channel statistics. To increase retinal feature visibility and image consistency, a specialized 13-stage preprocessing pipeline was used, which included noise reduction, illumination normalization, border trimming, and contrast enhancement. Data augmentation was applied exclusively

to the training set to improve generalization and reduce overfitting, using medically appropriate transformations such as horizontal flipping, small-angle rotations, affine transformations, colour jittering, and Gaussian blur.

C. Model Architectures

EfficientNet-B3 (Baseline): Our shared baseline uses compound-scaled EfficientNet-B3 pre-trained on ImageNet. The final classification head was replaced with a dropout layer (p=0.3) followed by a fully connected layer mapping to 5 classes.

ViT-B/16 (Mahad): The ViT-B/16 model, a vision transformer network, takes 16×16 image patches and is initialized on ImageNet-21K. It employs a two step fine tuning approach, sequentially training from the classification head to the entire network. The learning rates differ, with the classification head having a higher rate than the pre-trained encoders.

Swin Transformer (Ishaque): Swin-Base with shifted window attention, pre-trained on ImageNet-22k. The hierarchical feature extraction enables multi-scale representation learning, which is beneficial for detecting DR lesions of varying sizes.

DeiT III (Aneel): Data-efficient Image Transformer trained with an improved training recipe including 3-Augment, Simple Random Cropping, and grayscale augmentation. We used the base variant pre-trained on ImageNet-1k.

DINOv2 ViT-B/14 (Priyanka): Meta's self-supervised vision transformer model, trained using self-distillation without labeling, achieves high-resolution feature maps with a patch size of 14×14, unlike ViT-B/16.

MobileNetV3-Large (Nakul): MobileNetV3-Large, an efficient CNN architecture derived via neural architecture search, is crucial for computational efficiency analysis despite having fewer parameters.

D. Training Configuration

All models were trained on Visual Studio with the NVIDIA RTX A4000 using PyTorch. AdamW was the main optimiser, with cosine annealing learning rate scheduling where appropriate. Early stopping reduced overfitting, monitoring validation performance across loss, macro F1-score, and quadratic weighted kappa. The best model checkpoint was selected based on validation performance, ensuring final testing with the strongest generalising model. To mitigate the severe class imbalance present in the EyePACS dataset, multiple balancing strategies were investigated throughout the experiments, including weighted random sampling, class-weighted loss functions, focal loss, and clipped class weighting.

Parameter	Value
Optimiser	AdamW
Initial LR (head)	1-e4
LR (backbone)	1-e5
Weight decay	3-e5
Batch size	16
Epochs (max)	25-30
Image size	224×224
Dropout	0.3
LR scheduler	Cosine annealing
Early stopping patience	5

TABLE II. Common training hyperparameters across models.

IV. EXPERIMENTS AND RESULTS

A. Evaluation Metrics

Four complementing metrics are used to assess model performance: per-class precision and recall, quadratic weighted Cohen's kappa (the usual metric for ordinal DR grading), macro-averaged F1 score (to adjust for class imbalance), and overall classification accuracy. Because it penalizes predictions that deviate further from the true grade and reflects clinical relevance, the quadratic weighted kappa is especially significant.

B. Hyperparameter Sensitivity

Throughout the primary model studies, hyperparameter tuning was carried out by modifying the learning rate, weight decay, dropout, batch size, augmentation strength, and imbalance handling technique. For transformer-based models that had already been trained, learning rate was essential. Increased backbone learning rates resulted in decreased generalization and erratic validation performance, indicating a loss of valuable pre-trained representations. Classifier heads underwent stronger adaptation, while pre-trained backbones were fine-tuned at lower learning rates.

C. Model Comparison

The comparative outcomes of all six architectures assessed on the held-out validation set are shown in Table III. To guarantee a fair comparison, all models were trained under identical experimental settings.

The results reveal a clear hierarchy among the tested

Model	QWK	Macro F1	No DR	Mild	Moderate	Severe	Prolif.
EfficientNet-B4	0.5200	0.3965	0.7750	0.1721	0.3567	0.2893	0.3894
DINOv2	0.6947	0.5334	0.8898	0.1702	0.5500	0.4508	0.6064
SwinV2-Small	0.6778	0.5207	0.8392	0.1911	0.4742	0.4671	0.6321
DeiT3-B16	0.6470	0.5209	0.8580	0.1790	0.5029	0.4309	0.6337
MobileNetV3	0.6338	0.4695	0.8872	0.0661	0.4929	0.3874	0.5140
ViT-B16	0.5612	0.4553	0.8011	0.1433	0.4214	0.4422	0.5136

TABLE III. Model performance comparison on test set.

architectures. DINOv2 (ViT-B/14) achieved the best results in all measures, with an accuracy of 0.7835%, F1 score of 0.5334, and Kappa of 0.6947. Because Swin Transformer can use hierarchical attention to capture multi level information, it came in second. Among the CNNs, EfficientNet-B3 performed better than MobileNetV3-Large.

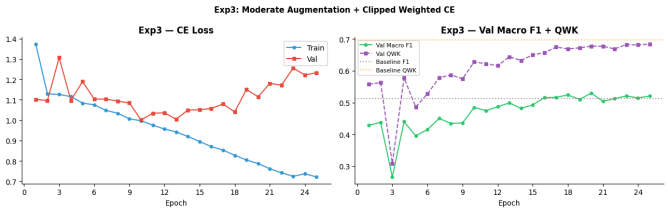


Fig. 1. Training and validation loss/accuracy curves for the best model.

C. Confusion Matrix Analysis

Our top performing model's confusion matrix is shown in Fig. 2. Because there are many training data in class 0 (No DR), the model gets the maximum accuracy on this category. The most difficult classes are Classes 1 (Mild) and 3 (Severe), where adjacent severity levels are frequently misclassified. Since the distinctions between successive DR grades are intrinsically unclear, this trend is clinically predicted.

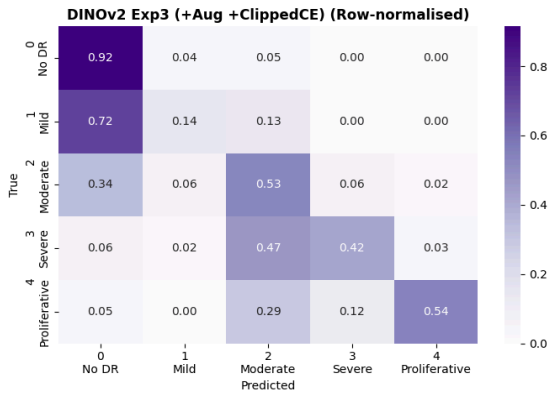


Fig. 2. Confusion matrix for the DINOv2 model.

D. Per-Class Performance

The model performance variation across diabetic retinopathy severity levels is shown by the per class F1-scores in Figure 3. Due to more training samples and more robust learning, Class 0 (No DR) has the greatest F1-score. Due to class imbalance and minute visual differences, Class 1 (Mild) has the lowest F1-score, indicating that the model has difficulty differentiating between early-stage DR and normal images. This issue has persisted in all models. Classes 3 (Severe) and 4 (Proliferative DR) have moderate F1-scores despite fewer samples, likely due to more visually distinctive pathological features.

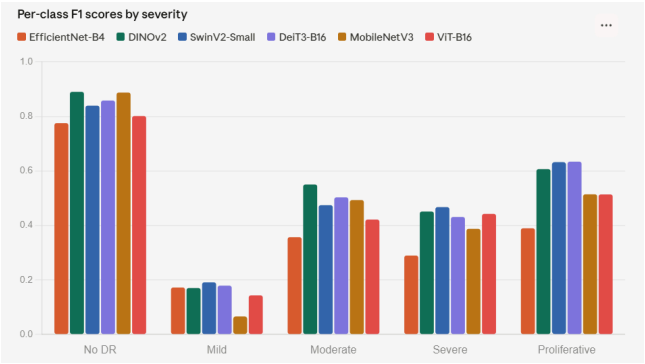
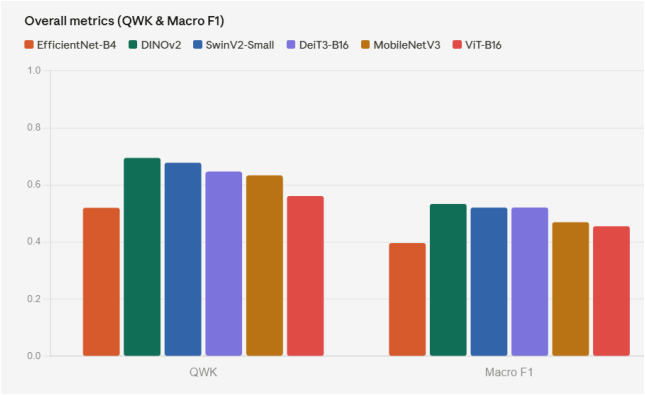


Fig. 3. Per-class F1 scores across models

E. Model Comparison Visualisation

A visual comparison of all six models across the three main evaluation metrics is shown in Fig. 4. The CNN baselines are regularly outperformed by the transformer-based models, with DINOv2 and Swin Transformer winning in every metric. Self-supervised (DINOv2) and supervised (ViT-B/16, DeiT III) converters



perform significantly differently.

Fig. 4. Comparison of all models across kappa and macro F1.

F. ROC Analysis

Fig. 5 shows the receiver operating characteristic (ROC) curves for each DR class. The model achieves the highest AUC for class 0 (No DR) and class 4 (Proliferative DR), which have the most distinctive visual characteristics. The one-vs-rest ROC analysis confirms that the model maintains strong discriminative ability across all classes.

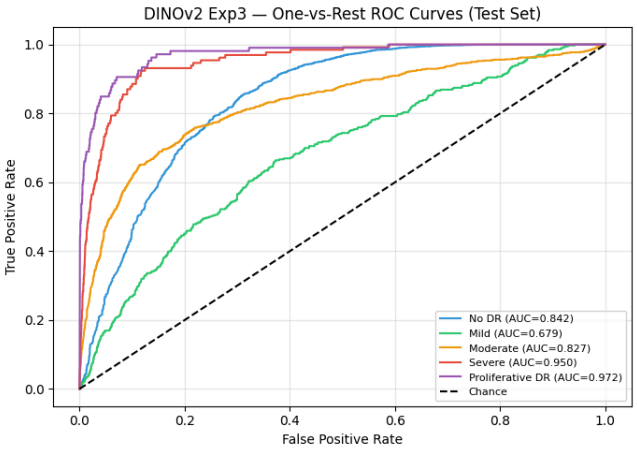


Fig. 5. ROC curves for each DR class (best model).

H. Model Explainability

We implemented Attention Rollout to SwinV2-Small and Gradient-weighted Class Activation Mapping (Grad-CAM) to EfficientNet-B3 in order to comprehend model decisions beyond aggregate metrics. The clinical reality that advanced DR exhibits clear pathological symptoms is reflected in the most confident right predictions, which show a monotonic relationship between disease severity and model confidence, ranging from 74.1% for No DR to 99.9% for Proliferative. Grad-CAM heatmaps verify that in mild and healthy situations In contrast to severe and proliferative grades, the model concentrates explicitly on hemorrhage clusters and neovascularization locations that meet clinical diagnostic criteria. The most significant failure mode is Grade 1 to Grade 0 misclassification, where Grad-CAM indicates that the model is focusing on the peripapillary vasculature and optic disc instead of identifying the minute microaneurysms that differentiate Mild DR, which explains the nearly 0% Mild F1 found in every trial. Preprocessing artifacts like circular mask boundaries and edge color inconsistencies appear to be the cause of a tiny proportion of high-confidence catastrophic errors, when the predicted grade deviates from the true label by two or more levels, rather than actual pathology. These results confirm that the models learn clinically relevant spatial features for moderate-to-severe DR, but they are still constrained by the sensitivity to imaging artifacts and the inherent ambiguity of early disease. These findings point to targeted augmentation and ordinal-aware loss functions as important areas for improvement.

V. CONCLUSION AND FUTURE WORK

Five deep learning architectures for automated diabetic retinopathy grading are thoroughly compared in this work. Our tests show that vision transformers outperform CNN baselines in this difficult medical picture classification task, especially those with self-supervised pre-training (DINOv2). Across all assessed metrics, the DINOv2 ViT-B/14 model produced the best results, indicating that self-supervised representations are more successful at capturing medically relevant characteristics than supervised pre-training on natural images.

Conclusions: Transformer based models outperform CNNs in DR classification. Self-supervised pre-training (DINOv2) is superior to supervised pre-training (ViT, DeiT III). Swin Transformer's hierarchical attention is beneficial for multi scale analysis of retinal diseases. Class imbalance persists, with mild DR being particularly challenging to classify.

Future directions may include using ensemble techniques to leverage multiple models, developing novel augmentation techniques for fundus images, using ordinal regression based on problem formulation, and utilizing larger pre-training datasets like EyePACS and UK Biobank. Incorporating visualizations like attention maps can also be considered.

REFERENCES

[1] V. Gulshan et al., "Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs," *JAMA*, vol. 316, no. 22, pp. 2402–2410, 2016.

[2] R. Gargeya and T. Leng, "Automated identification of diabetic retinopathy using deep learning," *Ophthalmology*, vol. 124, no. 7, pp. 962–969, 2017.

[3] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," *ICLR*, 2021.

[4] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," *ICCV*, pp. 10012–10022, 2021.

[5] H. Touvron et al., "DeiT III: Revenge of the ViT," *ECCV*, pp. 516–533, 2022.

[6] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," *TMLR*, 2024.

[7] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," *ICML*, pp. 6105–6114, 2019. [8] A. Howard et al., "Searching for MobileNetV3," *ICCV*, pp. 1314–1324, 2019.

[9] K. He et al., "Deep residual learning for image recognition," *CVPR*, pp. 770–778, 2016.

[10] B. Zhou et al., "Places: An image database for deep scene understanding," *arXiv:1610.02055*, 2016.

Appendix A: Datapreprocessing Visualisations

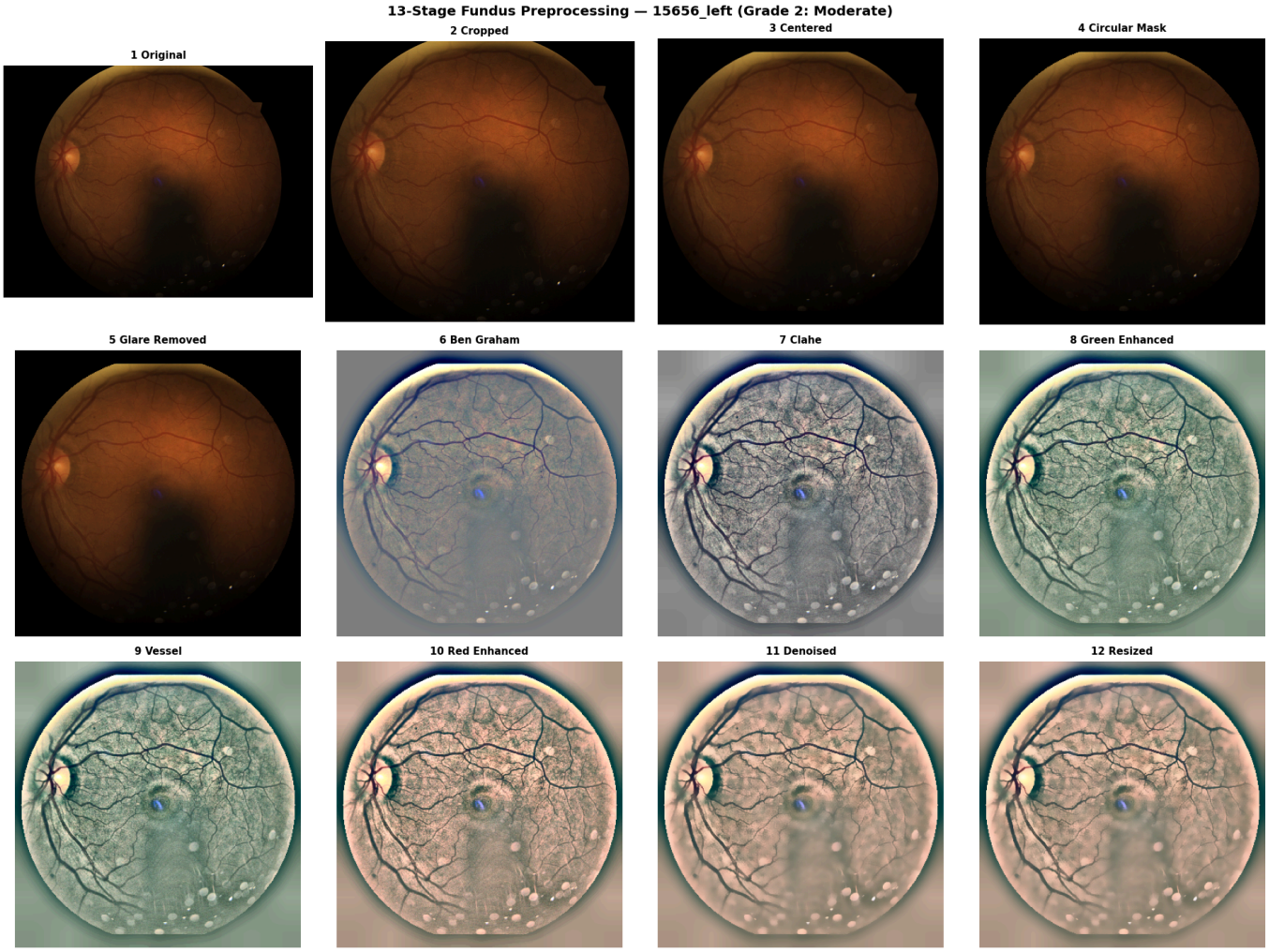


Figure A1: Overview of the 13-stage retinal image preprocessing pipeline applied to all fundus photographs, including border cropping, circular mask detection, contrast-limited adaptive histogram equalisation (CLAHE), illumination normalisation, and noise reduction, standardising image quality across the diverse imaging conditions present in the EyePACS dataset.

Appendix B: Explainability Visualisations

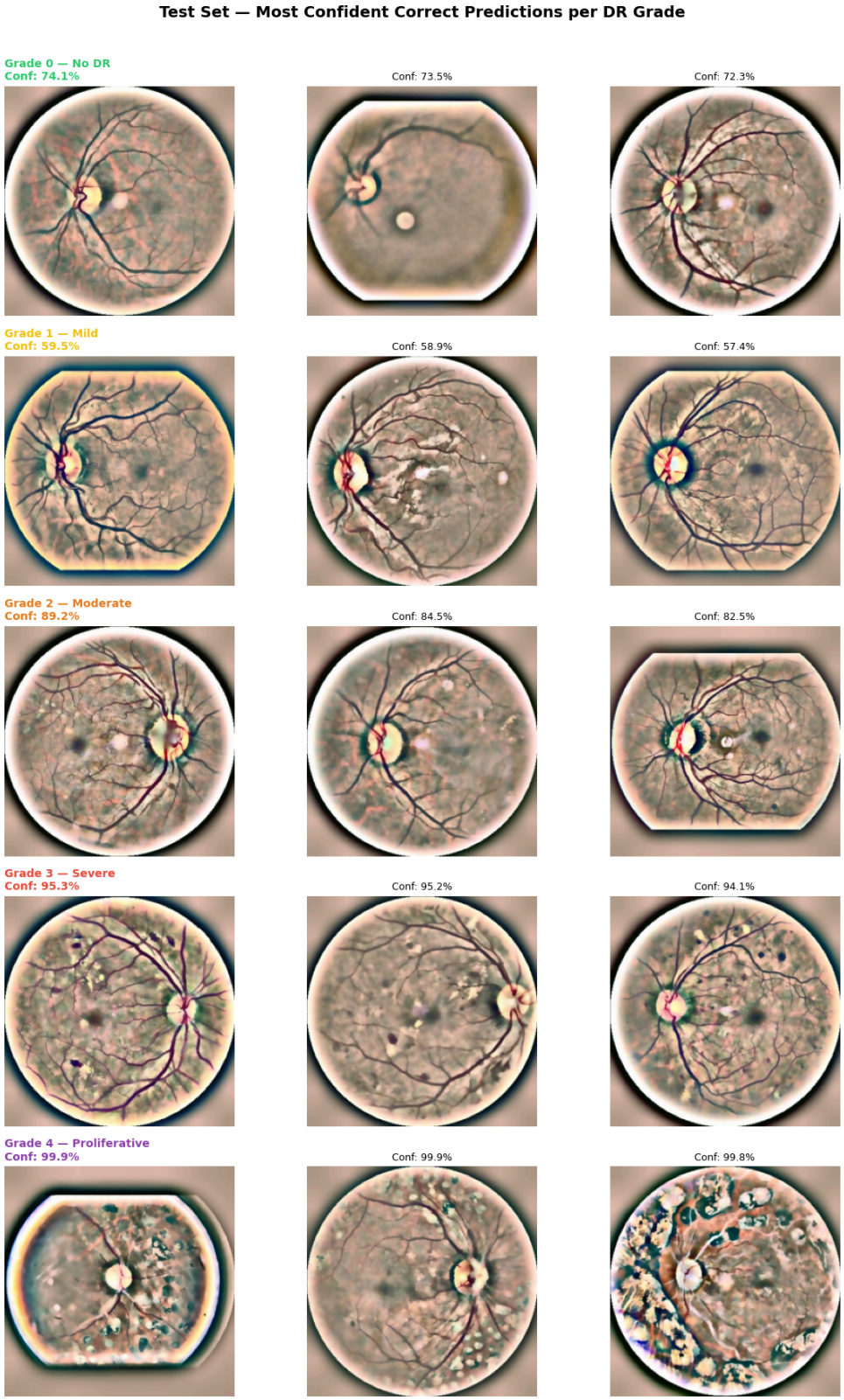


Figure B1 : Grade 0 peaks at 74.1% and Grade 4 reaches 99.9%, indicating that the algorithm finds advanced pathology far easier to classify than subtle early-stage disease. The model's confidence improves monotonically with severity.

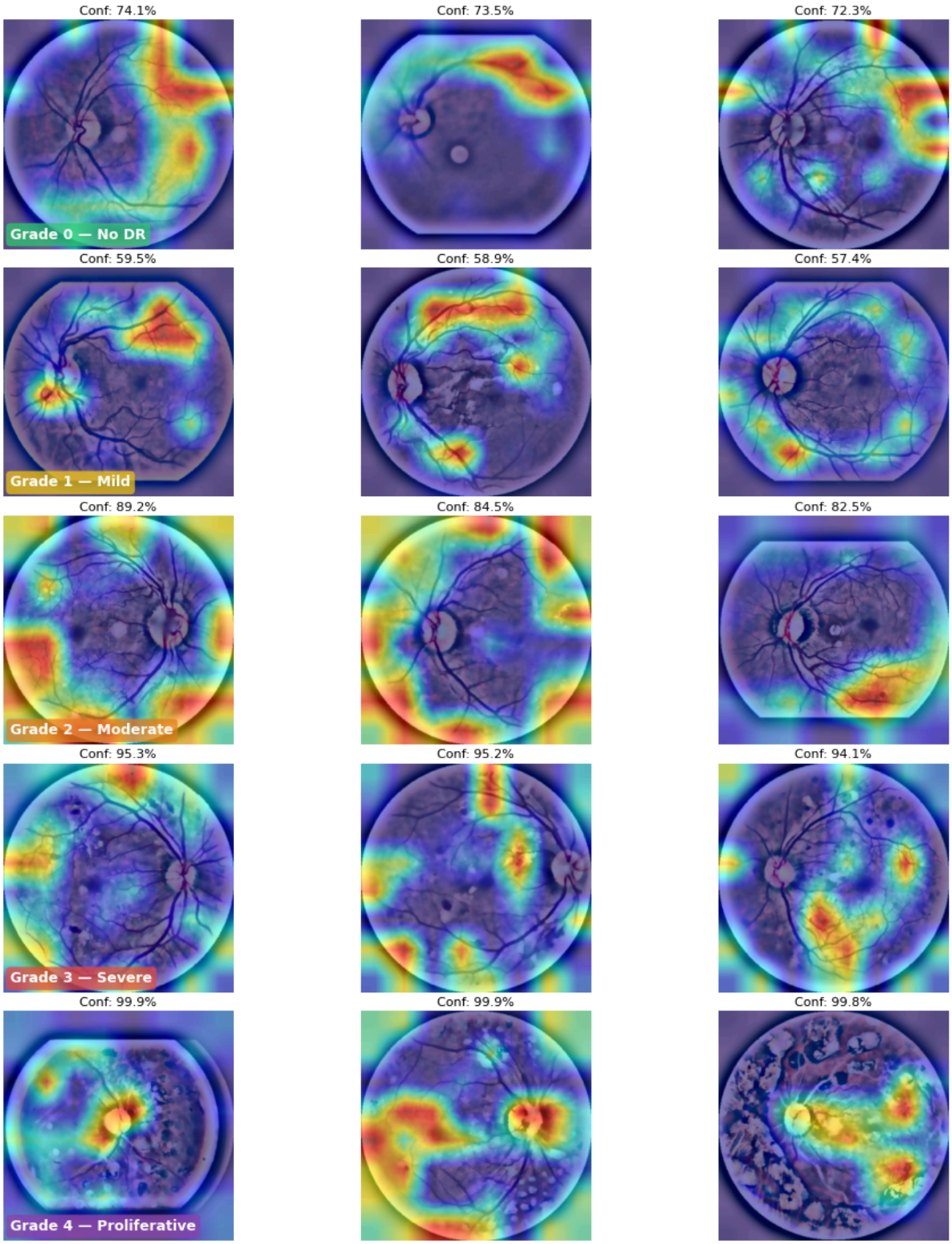


Figure B2 : Grad-CAM heatmaps for each of the five grades' most accurate and confident predictions. For No DR and Mild cases, activation spreads diffusely across the retina. Hemorrhage is the main focus for severe and proliferative grades.clusters and neovascularisation, consistent with clinical diagnostic features.

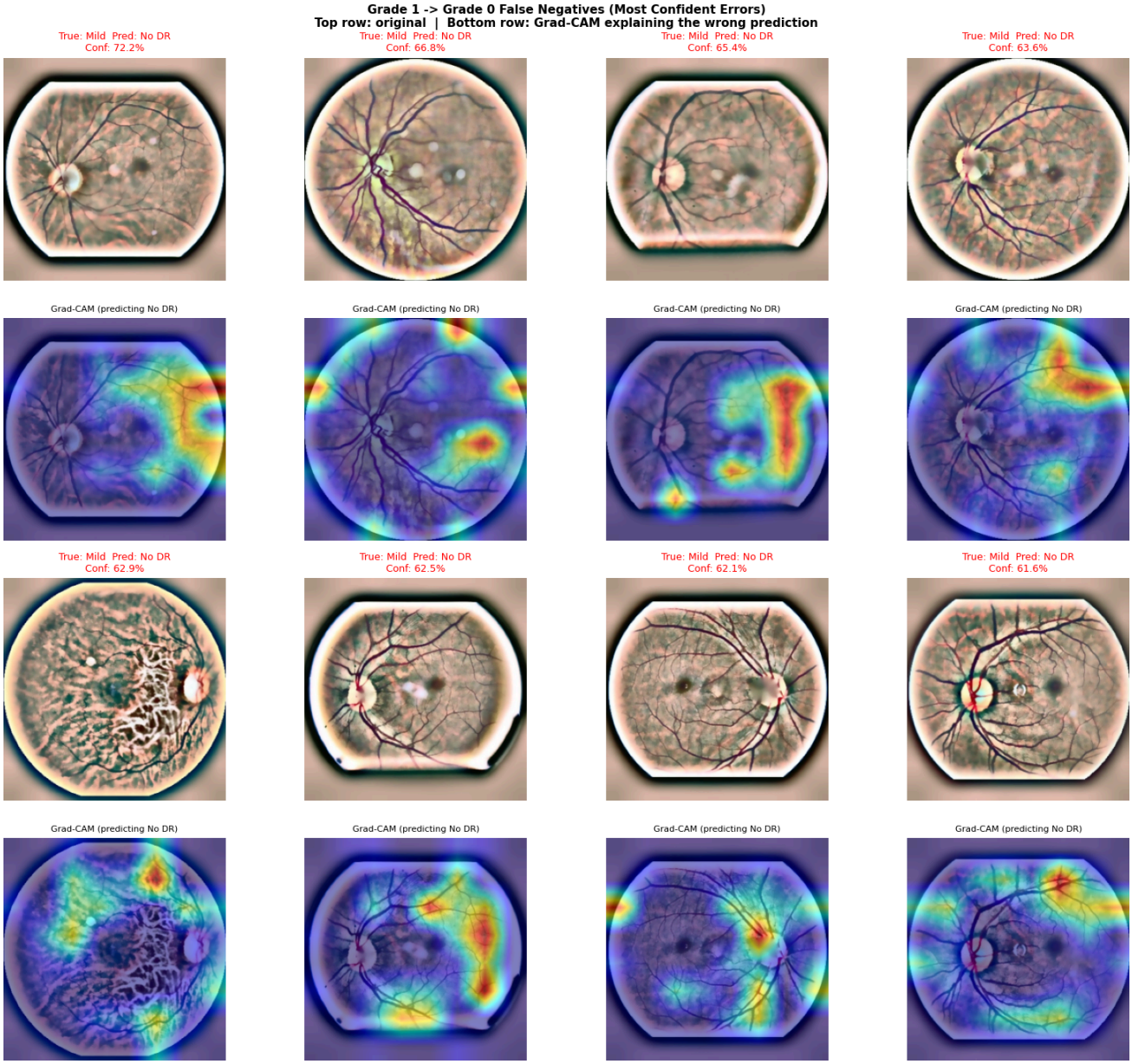


Figure B3: The most certain Grade 1→0 false negatives were analyzed using Grad-CAM. Instead of identifying the tiny microaneurysms that characterize Mild DR, the model focuses on the optic disc and main vasculature, the same areas used for healthy predictions, which explains the consistently low Mild F1 across all models.

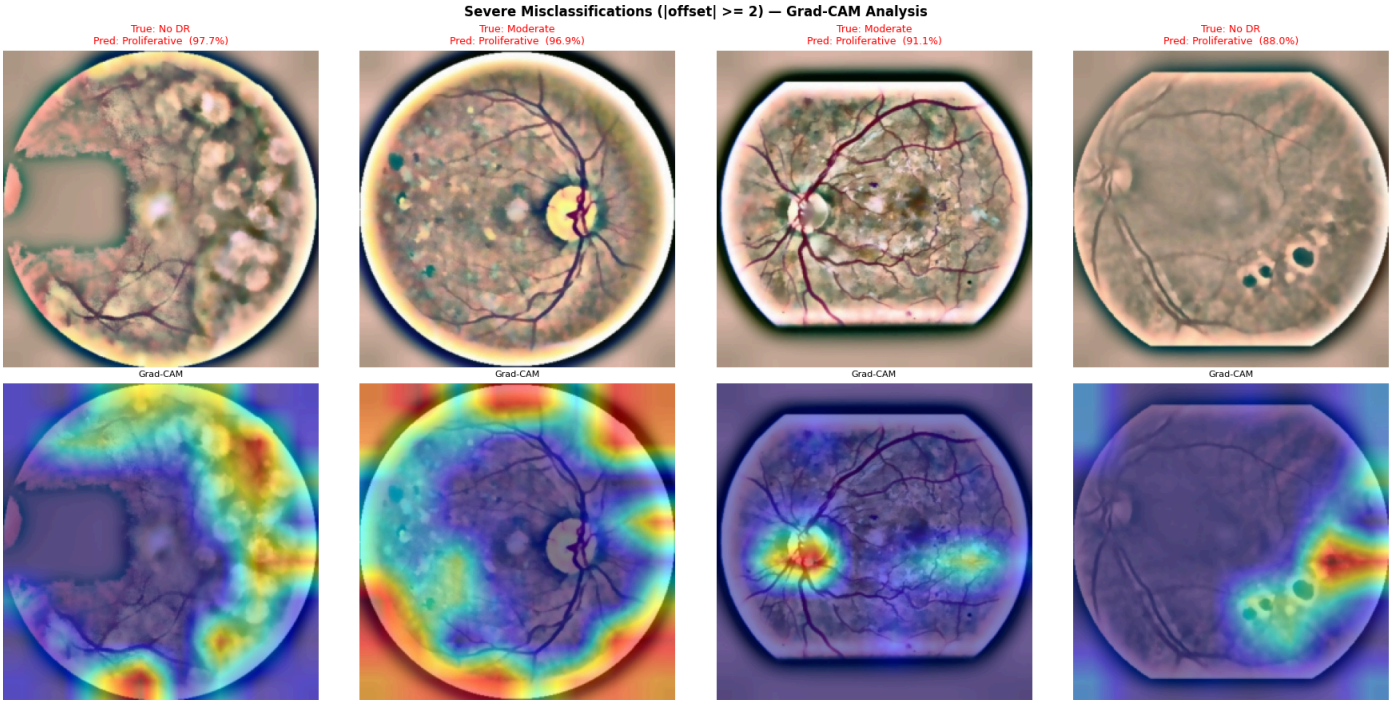


Figure B4: Severe misclassifications with Grad-CAM overlays (|offset| ≥ 2). The model responds to preprocessing artifacts such circular mask boundaries rather than actual retinal pathology in a number of high confidence errors, suggesting that data quality plays a role in catastrophic misclassification.

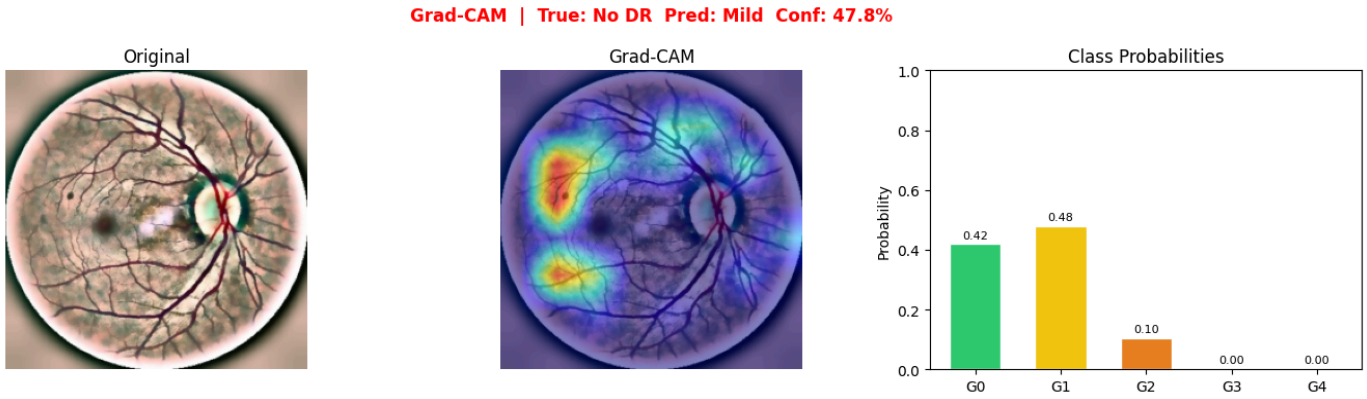


Figure B5: A Grade 0 (No DR) image was incorrectly identified as Mild DR at 47.8% confidence by Grad-CAM analysis. The concentration on the optic disc region and the nearly similar class probabilities (G0: 0.42, G1: 0.48) highlight the inherent ambiguity at the healthy mild boundary, where the model lacks discriminative features to reliably separate the two classes.